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The HGC³AE² Reference Architecture: A Cross-Paper Synthesis of the Mission-Ready AI Decalogy

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Abstract

The Mission-Ready AI Decalogy has, across its first nine papers, established the architecture of governed agentic operation. The HGC³AE² framework names the structural commitments — human-governed authority, curated context, agentially engineered execution, runtime verification — that distinguish trustworthy operation from capable but ungoverned operation. The Epistemic Constraints paper situates these commitments within the epistemic conditions agentic systems must satisfy. The Edge AI Doctrine specifies the seven-layer deployment frame. Alistair Prime in a Box defines the five pillars of the DDIL-tolerant cognitive node. The Skipjack Protocol formalizes the runtime enforcement layer. The Operating Model translates the framework into Scrum-extended human–AI team structure. The Agentic Substrate names the persona, runtime, persistence, and observability planes the system runs on. The Comprehension Instrument specifies the self-knowledge functions the system must run continuously. The Operational Plan empirically demonstrates the architecture in production federation.

This tenth paper is not a tenth claim. It is the reference architecture — the synthesis artifact that names the nine prior papers as components of a single coherent system, specifies their interfaces, and provides the cross-reference matrix by which any element of the architecture can be traced to its originating paper.

The paper proceeds as a specification document rather than as an argumentative whitepaper. Section 2 walks the architecture's components from foundation to capstone, with cross-references to the originating paper for each. Section 3 presents the formal cross-reference matrix. Section 4 specifies the conformance criteria an organization must demonstrate to claim Mission-Ready architecture conformance. Section 5 anchors the specification in the empirical evidence drawn from the heterogeneous federation documented in the Operational Plan. Section 6 returns the analysis to the HGC³AE² framework and hands off to the Managing Agentics Ops Decalogy and the Intent Horizon Series.

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ABSTRACT

The Mission-Ready AI Decalogy has, across its first nine papers, established the architecture of governed agentic operation. The HGC³AE² framework names the structural commitments — human-governed authority, curated context, agentially engineered execution, runtime verification — that distinguish trustworthy operation from capable but ungoverned operation.¹ The Epistemic Constraints paper situates these commitments within the epistemic conditions agentic systems must satisfy.² The Edge AI Doctrine specifies the seven-layer deployment frame.³ Alistair Prime in a Box defines the five pillars of the DDIL-tolerant cognitive node.⁴ The Skipjack Protocol formalizes the runtime enforcement layer.⁵ The Operating Model translates the framework into Scrum-extended human–AI team structure.⁶ The Agentic Substrate names the persona, runtime, persistence, and observability planes the system runs on.⁷ The Comprehension Instrument specifies the self-knowledge functions the system must run continuously.⁸ The Operational Plan empirically demonstrates the architecture in production federation.⁹

This tenth paper is not a tenth claim. It is the **reference architecture** — the synthesis artifact that names the nine prior papers as components of a single coherent system, specifies their interfaces, and provides the cross-reference matrix by which any element of the architecture can be traced to its originating paper. The reader's question — *can I build this?* — has, after this paper, the same form as the analogous question for any reference architecture in any technical discipline: *follow the specification, trace each element to its source paper, satisfy the conformance criteria, and the architecture has been built.*

The paper proceeds as a specification document rather than as an argumentative whitepaper. Section 2 walks the architecture's components from foundation to capstone, with cross-references to the originating paper for each. Section 3 presents the formal cross-

1 Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), Non Sequitur Publishing. <https://zenodo.org/records/19869285>.

2 Justin H. Kuiper, CISSP, *Epistemic Constraints* (v1.0-preprint, 2026), Non Sequitur Publishing. <https://zenodo.org/records/19869287>.

3 Justin H. Kuiper, CISSP, *Edge AI Doctrine* (v1.0-preprint, 2026), Non Sequitur Publishing. <https://zenodo.org/records/19869289>.

4 Justin H. Kuiper, CISSP, *Alistair Prime in a Box* (v1.0-preprint, 2026), Non Sequitur Publishing. <https://zenodo.org/records/19869307>.

5 Justin H. Kuiper, CISSP, *The Skipjack Protocol* (v1.0-preprint, 2026), Non Sequitur Publishing. <https://zenodo.org/records/19869293>.

6 Justin H. Kuiper, CISSP, *The Operating Model for Agentic Teams* (v1.0-preprint, 2026), Non Sequitur Publishing. <https://zenodo.org/records/19869313>.

7 Justin H. Kuiper, CISSP, *The Agentic Substrate* (v1.0-preprint, 2026), Non Sequitur Publishing. <https://zenodo.org/records/19869291>.

8 Forthcoming MR-P8 *The Comprehension Instrument for Agentic Systems* (v1.0-preprint, in concurrent drafting; yks-build#98).

9 Forthcoming MR-P9 *Operational Plan: A Heterogeneous Federation as Proof of the Mission-Ready Architecture* (v1.0-preprint, in concurrent drafting; yks-build#99).



reference matrix. Section 4 specifies the conformance criteria an organization must demonstrate to claim Mission-Ready architecture conformance. Section 5 anchors the specification in the empirical evidence drawn from the heterogeneous federation documented in the Operational Plan. Section 6 returns the analysis to the HGC³AE² framework and hands off to the Managing Agentics Ops Decalogy (which extends the architecture into operational depth) and the Intent Horizon Series (which extends it across operational time).

The central claim of this paper is that the Mission-Ready architecture is now sufficiently specified to be implemented, audited, and adapted by organizations independent of the Non Sequitur reference implementation. Capability without architecture invites confident misalignment. The reference architecture documented here is the architecture that, when applied, prevents confident misalignment from being the default operating state of agentic systems.

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1. INTRODUCTION

Reference architectures occupy a particular position in technical disciplines. They are not new claims; they are integrations of prior claims into specifications detailed enough to be implemented, audited, and adapted by parties other than the architecture's authors. The TOGAF framework occupies this position for enterprise architecture; SABSA for security; the NIST Cybersecurity Framework for organizational cybersecurity posture.¹⁰¹¹¹² The peer-reviewed software architecture literature has developed a substantial body of methodology around what reference architectures are, how they differ from product-line architectures, and what makes one empirically grounded rather than merely aspirational. Galster and Avgeriou's empirical-grounding proposal¹³ and Angelov, Grefen, and Greefhorst's framework for analysis and design of software reference architectures¹⁴ together specify the methodological tests this paper applies. Cloutier and colleagues' systems-engineering treatment of the concept of reference

10 The Open Group, *The TOGAF Standard*, 10th edition (Reading, MA: The Open Group, 2022).

11 John Sherwood, Andrew Clark, and David Lynas, *Enterprise Security Architecture: A Business-Driven Approach* (Boca Raton, FL: CRC Press, 2005).

12 National Institute of Standards and Technology, *Cybersecurity Framework Version 2.0*, NIST CSWP 29 (Gaithersburg, MD, 2024), <https://doi.org/10.6028/NIST.CSWP.29>.

13 Matthias Galster and Paris Avgeriou, "Empirically-Grounded Reference Architectures: A Proposal," in *Proceedings of the Joint ACM SIGSOFT Conference QoSA and ACM SIGSOFT Symposium ISARCS on Quality of Software Architectures (QoSA) and Architecting Critical Systems (ISARCS)* (New York: ACM, 2011), 153–158, <https://doi.org/10.1145/2000259.2000285>.



architectures provides the broader systems-engineering frame.¹⁵ The HGC³AE² Reference Architecture documented in this paper is the equivalent integration for governed agentic operation.

The reference-architecture treatment differs from the argumentative whitepaper treatment that characterizes the preceding nine papers in three respects. First, the burden of justification has been shifted: the nine prior papers carry the burden of establishing *why* each architectural commitment matters; the present paper assumes those justifications and carries instead the burden of specifying *how* the commitments compose into a single system. The software-architecture literature has developed canonical structures for this kind of specification document, exemplified by Bass, Clements, and Kazman's treatment of how architectures are described, analyzed, and evaluated.¹⁶ Second, the structure of the document is specification-shaped rather than argument-shaped: each section names a layer, identifies its interfaces, specifies its conformance criteria, and cross-references its originating paper. Third, the page-floor and citation-density commitments that govern preceding papers in the series are observed differently: the cross-reference matrix in Section 3 carries citation density not by footnote count but by formal element-to-paper mapping.

The architecture being specified is composed of seven layers, organized from foundation to application:

Foundation Layer (P1, P2, P3): the HGC³AE² framework, the epistemic constraints, and the edge AI doctrine that jointly establish what governed agentic operation must satisfy.

Node Layer (P4): the five pillars of the DDIL-tolerant cognitive node, the unit of agentic execution.

Operating Layer (P5, P6): the Skipjack Protocol and the Scrum-extended operating model that govern runtime operation.

Substrate Layer (P7): the persona, runtime, persistence, and observability planes that the system runs on.

Cognitive Layer (P8): the Comprehension Instrument that maintains the system's self-knowledge.

Operational Layer (P9): the federation pattern and empirical validation drawn from production operation.

Capstone Layer (P10): the present paper, which integrates the layers above into the reference architecture.

Each layer interfaces with its neighbors according to the protocols and contracts specified by the originating paper. The integration the present paper provides is not a re-derivation of these interfaces; it is a unified specification of how they compose. Section 2 walks the layers in order.

14 Samuil Angelov, Paul Grefen, and Danny Greefhorst, "A Framework for Analysis and Design of Software Reference Architectures," *Information and Software Technology* 54, no. 4 (2012): 417–431, <https://doi.org/10.1016/j.infsof.2011.11.009>.

15 Robert Cloutier, Gerrit Muller, Dinesh Verma, Roshanak Nilchiani, Eirik Hole, and Mary Bone, "The Concept of Reference Architectures," *Systems Engineering* 13, no. 1 (2010): 14–27, <https://doi.org/10.1002/sys.20129>.

16 Len Bass, Paul Clements, and Rick Kazman, *Software Architecture in Practice*, 3rd ed. (Boston: Addison-Wesley, 2012).



The remainder of this paper proceeds in the canonical Mission-Ready shape, adapted for capstone structure. Section 2 walks the architecture's components with cross-references. Section 3 presents the cross-reference matrix. Section 4 specifies the conformance criteria. Section 5 anchors the architecture in the operational evidence. Section 6 returns to HGC³AE² and closes the Decalogy.

2. THE REFERENCE ARCHITECTURE'S COMPONENTS

The architecture's seven layers are walked here in foundation-to-capstone order. Each subsection identifies the layer, names its components, references the originating paper, and specifies the layer's interfaces with its neighbors.

2.1 FOUNDATION LAYER

The **Foundation Layer** comprises three papers that jointly establish the conditions under which governed agentic operation is meaningful.

The **HGC³AE² framework** itself (P1) names four structural commitments — Human-governed (H), Governance-with-Curated-Context (G), and Agentially-Engineered-and-Executed (AE), with the exponent capturing the closure of the loop — that together distinguish trustworthy operation from capable but ungoverned operation.¹⁷ The framework's central claim is that scalable and trustworthy agentic deployment depends less on maximizing autonomy than on embedding enforceable mechanisms that keep human intent, contextual truth, and system execution synchronized over time. The reference architecture inherits this claim as its top-level axiom: every other layer is justified, in part, by how it serves the HGC³AE² commitments.

The **Epistemic Constraints** paper (P2) develops the conditions under which agentic systems can reliably know what they know.¹⁸ The paper identifies epistemic degradation as a structural failure mode and specifies the architectural responses required to prevent it. The reference architecture inherits these constraints as runtime invariants that the substrate (P7), the comprehension instrument (P8), and the operational layer (P5) jointly enforce.

The **Edge AI Doctrine** (P3) specifies the seven-layer edge model and the deployment doctrine for agentic systems operating outside hyperscaler-default conditions.¹⁹ The doctrine establishes which workloads belong forward (closer to the operator and the data), which belong rearward (closer to centralized governance), and how the boundary between forward and rearward is managed across the deployment lifecycle. The reference architecture inherits the seven-layer model as the deployment frame; the federation pattern documented in P9 is one concrete instance of the doctrine's principles in practice.

Foundation Layer outputs flow downward into the Node Layer (P4 inherits the doctrine), the Operating Layer (P5 and P6 inherit the framework and epistemic constraints), and the Cognitive Layer (P8 inherits all three).

17 Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), Non Sequitur Publishing. <https://zenodo.org/records/19869285>.

18 Justin H. Kuiper, CISSP, *Epistemic Constraints* (v1.0-preprint, 2026), Non Sequitur Publishing. <https://zenodo.org/records/19869287>.

19 Justin H. Kuiper, CISSP, *Edge AI Doctrine* (v1.0-preprint, 2026), Non Sequitur Publishing. <https://zenodo.org/records/19869289>.



2.2 NODE LAYER

The **Node Layer** comprises the Alistair Prime in a Box paper (P4), which specifies the five pillars of the DDIL-tolerant cognitive node.²⁰ The pillars are: (1) local inference capability, (2) persistent memory, (3) agentic orchestration, (4) governance enforcement, and (5) execution capability. Each pillar is independently necessary; the node is the unit composed of all five pillars together.

The Node Layer is the *unit* of agentic execution in the reference architecture. Other layers compose nodes (the federation pattern in P9 composes multiple nodes across substrates), supervise nodes (the operating layer in P5/P6), or run on nodes (the substrate in P7 specifies what runs *on* a node). The reference architecture's portability claim — that an organization adopting the architecture can do so on commercially available infrastructure — depends on the Node Layer being substrate-agnostic, which is what the five-pillar specification provides.

Node Layer outputs flow upward to the Operating Layer (which orchestrates nodes), the Substrate Layer (which runs nodes), and the Cognitive Layer (which runs *in* nodes).

2.3 OPERATING LAYER

The **Operating Layer** comprises the Skipjack Protocol (P5) and the Operating Model for Agentic Teams (P6). Together they specify the runtime governance under which nodes operate and the human–AI team structure under which the system as a whole is operated.

The **Skipjack Protocol** (P5) names four enforcement mechanisms — context integrity and epistemic continuity; structured task decomposition and routing; human-in-the-loop control points; iterative feedback loops — that together translate the HGC³AE² framework's structural commitments into runtime enforcement.²¹ The protocol's principles (live validation of dynamic state; no silent fallback; explicit uncertainty signaling; reconciliation checkpoints; governance-gated execution; runtime verification of design assumptions; emergent and iterative application) are the operational vocabulary of the architecture's runtime layer.

The **Operating Model** (P6) extends Scrum's authority structure into a Scrum-Skipjack remap suitable for hybrid human–AI teams.²² The Product Owner maps to a Governance Layer; the Scrum Master maps to an Orchestrator; the Development Team maps to an Agentic Execution Layer; the Backlog maps to persistent memory and task queues. The model preserves Scrum's defined authority structure while accounting for the participation of AI agents in execution.

Operating Layer outputs flow downward to the Node Layer (nodes execute under Skipjack), upward to the Cognitive Layer (the Comprehension Instrument couples to Skipjack mechanisms), and outward to the human operators who staff the Governance Layer.

²⁰ Justin H. Kuiper, CISSP, *Alistair Prime in a Box* (v1.0-preprint, 2026), Non Sequitur Publishing. <https://zenodo.org/records/19869307>.

²¹ Justin H. Kuiper, CISSP, *The Skipjack Protocol* (v1.0-preprint, 2026), Non Sequitur Publishing. <https://zenodo.org/records/19869293>.

²² Justin H. Kuiper, CISSP, *The Operating Model for Agentic Teams* (v1.0-preprint, 2026), Non Sequitur Publishing. <https://zenodo.org/records/19869313>.



2.4 SUBSTRATE LAYER

The **Substrate Layer** comprises the Agentic Substrate paper (P7), which specifies four planes — persona, runtime, persistence, and observability — that the system operates on.²³ The persona plane represents the agentic identities the system addresses (Taskmaster, Clive, Edna, Klaus, Stackrat, etc.). The runtime plane represents the compute and accelerator capacity that personas execute on. The persistence plane represents the durable state — memory, governance event log, intermediate artifacts — that survives across operational sessions. The observability plane represents the telemetry, logs, and metrics that operators consume.

The substrate is composable. The federation pattern documented in P9 is one composition of substrates (on-premises K3s + hyperscaler managed + sovereign bare-metal); other compositions are admissible by the reference architecture, subject to the conformance criteria specified in Section 4.

Substrate Layer outputs flow upward to the Node Layer (nodes run on the substrate) and to the Operating Layer (the Orchestrator routes across the substrate).

2.5 COGNITIVE LAYER

The **Cognitive Layer** comprises the Comprehension Instrument paper (P8), which specifies five functions — State Introspection, Intent Rehearsal, Envelope-Distance Estimation, Drift Detection, and Scope-Boundary Signaling — that the agentic system must run continuously to maintain operationally meaningful self-knowledge.²⁴ The functions are mutually reinforcing; the instrument is the architectural mechanism by which the system's runtime cognition couples to the surrounding governance machinery.

The Cognitive Layer is the runtime cognition that the reference architecture distinguishes from model capability. A sufficiently capable model is not, by virtue of capability alone, in possession of the comprehension functions; the functions are an architectural commitment that the system must structurally satisfy. Cognitive Layer outputs feed every Skipjack enforcement mechanism (P5) and inform every Governance Layer adjustment (P6).

2.6 OPERATIONAL LAYER

The **Operational Layer** comprises the Operational Plan paper (P9), which documents the heterogeneous federation as as-deployed proof of the architecture.²⁵ The federation — Omen on-premises + AWS managed + OCI sovereign bare-metal — demonstrates that the architecture is operationally viable today, at unit economics that single-hyperscaler deployment cannot match for the workload mix mission-ready agentic operation produces.

The Operational Layer is the empirical anchor for the reference architecture's claims. The architecture is not a future-tense proposal; it has been deployed and operated, with measurable outcomes documented in P9's cost ledger and governance event log.

²³ Justin H. Kuiper, CISSP, *The Agentic Substrate* (v1.0-preprint, 2026), Non Sequitur Publishing. <https://zenodo.org/records/19869291>.

²⁴ Forthcoming MR-P8 *The Comprehension Instrument for Agentic Systems* (v1.0-preprint, in concurrent drafting; yks-build#98).

²⁵ Forthcoming MR-P9 *Operational Plan: A Heterogeneous Federation as Proof of the Mission-Ready Architecture* (v1.0-preprint, in concurrent drafting; yks-build#99).



2.7 CAPSTONE LAYER

The **Capstone Layer** is the present paper. It does not introduce new architectural claims; it integrates the layers above into a single specification that organizations independent of the Non Sequitur reference implementation can adopt, audit, and adapt.

The capstone closes the Decalogy and hands off to the Managing Agentics Ops Decalogy (which develops the operational depth of the architecture) and the Intent Horizon Series (which develops the architecture's response to intent fidelity over operational time).







3. THE CROSS-REFERENCE MATRIX

The matrix below maps every architectural element of the reference architecture to its originating paper(s). The matrix is the formal specification artifact that allows an implementer to trace any architectural decision back to its source.

Architectural Element

Originating Paper

Section / Construct

Human authority structure (H of HGC³AE²)

P1

§3 (framework definition)

Curated Context with collaboration boundary (G/C)

P1

§3, §4 (collaboration as continuous process)

Agentially Engineered + Executed (AE²)

P1

§3, §5 (Skipjack as operational layer)

Confident misalignment failure mode

P1

§1, §2 (problem set)

Epistemic degradation as structural failure

P2

§1, §2

Seven-layer edge model

P3

§2 (the seven layers)

Forward / rearward deployment doctrine

P3

§3

Five pillars of the cognitive node

P4



§3 (pillar definitions)

Pre-authorized decision envelope

P4

Pillar 4 (governance)

Persistent memory pillar

P4

Pillar 2

Four Skipjack enforcement mechanisms

P5

§3

Live validation of dynamic state

P5

§3 (principle 1)

No silent fallback

P5

§3 (principle 2)

Explicit uncertainty signaling

P5

§3 (principle 3)

Reconciliation checkpoints

P5

§3 (principle 4)

Governance-gated execution

P5

§3 (principle 5)

Scrum-Skipjack role remap

P6

§2 (role remap table)

Governance Layer (PO remap)

P6

§2.3

Orchestrator (Scrum Master remap)

P6

§2.4

Agentic Execution Layer (Dev Team remap)



P6

§2.5

Persistent memory + task queues (Backlog remap)

P6

§2.6

Persona plane

P7

substrate spec

Runtime plane

P7

substrate spec

Persistence plane

P7

substrate spec

Observability plane

P7

substrate spec

State Introspection (S) function

P8

§3.1

Intent Rehearsal (I) function

P8

§3.2

Envelope-Distance Estimation (D) function

P8

§3.3

Drift Detection (T) function

P8

§3.4

Scope-Boundary Signaling (B) function

P8

§3.5

Comprehension-Skipjack coupling

P8

§5



Heterogeneous federation pattern

P9

§3 (federation as built)

Unit-economics ratio observation (~0.55)

P9

§4.4

Persistence-tier anchor on Omen

P9

§3.1

Cross-substrate Skipjack enforcement

P9

§5

Reference architecture integration

P10

this paper, §2

Conformance specification

P10

this paper, §4

Cross-reference matrix (this table)

P10

this paper, §3

The matrix is intentionally exhaustive. An architecture without traceability is not auditable; an architecture not auditable cannot be claimed conformant. Section 4 specifies what conformance requires.

4. CONFORMANCE SPECIFICATION

An organization may claim conformance to the HGC³AE² Reference Architecture by demonstrating, at minimum, the following:

4.1 FOUNDATION-LAYER CONFORMANCE

The organization must demonstrate that its agentic operation is governed by the HGC³AE² framework's four structural commitments.²⁶ Conformance evidence at this layer takes the form of:

A documented Governance Layer composed of identifiable humans with accountability for the agentic system's intent, constraints, and decision envelopes.

²⁶ Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), Non Sequitur Publishing. <https://zenodo.org/records/19869285>.



A documented curated-context tier distinct from raw model state, with explicit currency metadata.

A documented architecture for agentially engineered execution operating within governance-defined envelopes.

Evidence that the loop closes (the exponent) — that execution outputs inform Governance Layer envelope updates on a defined cadence.

4.2 NODE-LAYER CONFORMANCE

The organization must demonstrate that the unit of agentic execution satisfies the five-pillar specification of the cognitive node.²⁷ Conformance evidence:

Local inference capability (Pillar 1) — the node can perform inference without dependence on continuous external connectivity.

Persistent memory (Pillar 2) — the node maintains durable state across operational sessions.

Agentic orchestration (Pillar 3) — the node decomposes tasks and routes them under governance.

Governance enforcement (Pillar 4) — the node operates within a pre-authorized decision envelope and surfaces envelope-boundary events.

Execution capability (Pillar 5) — the node executes assigned tasks within scope.

4.3 OPERATING-LAYER CONFORMANCE

The organization must demonstrate Skipjack Protocol enforcement of all four mechanisms²⁸ and adoption of the Scrum-Skipjack role remap.²⁹ Conformance evidence:

Context integrity events logged at the runtime.

Structured task decomposition and routing visible in the orchestration plane.

HITL control points that fire on substantive operational transitions, not on fixed cadence alone.

Iterative feedback loops with documented closure (envelope changes traceable to feedback content).

Governance Layer / Orchestrator / Agentic Execution Layer roles identifiable in the operating organization.

²⁷ Justin H. Kuiper, CISSP, *Alistair Prime in a Box* (v1.0-preprint, 2026), Non Sequitur Publishing. <https://zenodo.org/records/19869307>.

²⁸ Justin H. Kuiper, CISSP, *The Skipjack Protocol* (v1.0-preprint, 2026), Non Sequitur Publishing. <https://zenodo.org/records/19869293>.

²⁹ Justin H. Kuiper, CISSP, *The Operating Model for Agentic Teams* (v1.0-preprint, 2026), Non Sequitur Publishing. <https://zenodo.org/records/19869313>.



4.4 SUBSTRATE-LAYER CONFORMANCE

The organization must demonstrate that the four planes — persona, runtime, persistence, observability — are operationally present and unified.³⁰ Conformance evidence:

A persona registry that enumerates the agentic identities the system addresses.

A documented runtime tier with capacity and routing semantics.

An authoritative persistence tier with explicit currency and consistency semantics.

A unified observability plane through which operators reason about the system.

4.5 COGNITIVE-LAYER CONFORMANCE

The organization must demonstrate that the five comprehension functions are architecturally present.³¹ Conformance evidence:

State Introspection produces accurate, currency-tagged self-reports.

Intent Rehearsal records and exposes the system's operative intent.

Envelope-Distance Estimation produces consumable distance/proximity signals.

Drift Detection emits structured drift events at runtime.

Scope-Boundary Signaling surfaces relevant comprehension state to the orchestrator and operator.

4.6 OPERATIONAL-LAYER CONFORMANCE

The organization should — though is not required to — demonstrate operational validation evidence comparable to that documented in P9.³² An architecture that has been deployed under load and operated successfully across a meaningful operating window has stronger conformance evidence than one that has only been specified.

4.7 AUDITABILITY CONFORMANCE

The organization must maintain the architectural traceability documented in §3's cross-reference matrix. Every architectural decision in the deployed system must be traceable to the originating paper of the reference architecture. The matrix is not a documentation convenience; it is the auditability instrument.

The auditability commitment is informed by peer-reviewed work on assurance cases — the safety-engineering literature's principal instrument for arguing, evidence in hand, that a system satisfies its claims under specified operating conditions. Kelly and Weaver's Goal Structuring Notation provides one canonical structure for such arguments;³³ the reference architecture's

30 Justin H. Kuiper, CISSP, *The Agentic Substrate* (v1.0-preprint, 2026), Non Sequitur Publishing. <https://zenodo.org/records/19869291>.

31 Forthcoming MR-P8 *The Comprehension Instrument for Agentic Systems* (v1.0-preprint, in concurrent drafting; yks-build#98).

32 Forthcoming MR-P9 *Operational Plan: A Heterogeneous Federation as Proof of the Mission-Ready Architecture* (v1.0-preprint, in concurrent drafting; yks-build#99).

33 Tim Kelly and Rob Weaver, "The Goal Structuring Notation: A Safety Argument Notation," in *Proceedings of the Dependable Systems and Networks 2004 Workshop on*



traceability matrix is functionally analogous, claiming a defended mapping between each architectural commitment and the empirical or specificational evidence that justifies it. The AI accountability literature has begun to converge on similar instruments: Raji and colleagues' end-to-end framework for internal algorithmic auditing specifies the documentation, traceability, and review checkpoints that operationally-significant AI systems must satisfy.³⁴ The cross-reference matrix in §3, together with the operational evidence in §5, constitutes the architecture's audit-trail object.

5. OPERATIONAL VALIDATION EVIDENCE

The Operational Plan paper (P9) documents the Non Sequitur Robo Stack federation as as-deployed evidence that the architecture is operationally viable.³⁵ The federation's empirical record — workload-class routing telemetry, governance event log, persistence-tier consistency record, cost-ledger compilation, failure-mode and recovery documentation — constitutes the canonical validation evidence for the reference architecture's claims.

Three findings from the operational record are load-bearing for the architecture's adoption case:

First, the architecture survives heterogeneity. The federation runs the same agentic operating model across three substrates with materially different operational properties; the unified operating layer, the authoritative persistence tier, and the unified observability plane preserve the architecture's commitments without requiring the substrates themselves to be homogenized.

Second, the architecture survives failure. Across the operating window, multiple substrate-level and substrate-routing failures occurred (accelerator allocation tightening, regional sovereignty events, persistence-tier consistency departures, governance envelope breaches caught by Skipjack mechanisms). In each case, the architecture's failure-isolation and recovery semantics held. The federation continued to operate; the agentic operating model continued to satisfy governance commitments; the persistent state survived the substrate transitions.

Third, the architecture survives the cost question. The federation operates the same workload at approximately 55% of the cost of an equivalent single-hyperscaler reference deployment. The savings are not uniform across workload classes; they are heavily concentrated on the steady-state inference, persistent memory, and governance overhead workloads that dominate the mission-ready agentic mix.³⁶ Adoption of the architecture is, for organizations operating at this workload mix, economically as well as architecturally favorable.

The operational validation evidence is not the architecture; it is one demonstration that the architecture works. Other organizations adopting the reference architecture will accumulate their

Assurance Cases (Florence, Italy: IEEE, 2004).

³⁴ Inioluwa Deborah Raji, Andrew Smart, Rebecca N. White, Margaret Mitchell, Timnit Gebru, Ben Hutchinson, Jamila Smith-Loud, Daniel Theron, and Parker Barnes, "Closing the AI Accountability Gap: Defining an End-to-End Framework for Internal Algorithmic Auditing," in *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency (FAT '20)** (New York: ACM, 2020), 33–44, <https://doi.org/10.1145/3351095.3372873>.

³⁵ Forthcoming MR-P9 *Operational Plan: A Heterogeneous Federation as Proof of the Mission-Ready Architecture* (v1.0-preprint, in concurrent drafting; yks-build#99).

³⁶ Forthcoming MAO-P3 *The Capex/Opex Inflection* (v0.1-seed, in concurrent drafting; yks-build#94).



own validation evidence on their own substrates. The reference architecture is portable in the sense that it does not require the Non Sequitur reference implementation specifically.

6. IMPLICATIONS AND CONCLUSION

The implications of the reference architecture extend across organizational governance, public administration, defense operations, and the broader question of how AI deployment should be governed in regulated environments. The architecture is not specific to any one jurisdiction or sector; it is specific to the structural problem of governed agentic operation, which is now a problem that every organization deploying agentic systems must address.³⁷³⁸³⁹ Recent AI ethics literature has converged on the need for ethics-and-governance instruments that are operationally instantiated rather than declared. Floridi and colleagues' AI4People framework identified the gap between ethical principles and operational practice as the principal obstacle to trustworthy AI deployment;⁴⁰ Mittelstadt and colleagues' map of the algorithmic ethics debate identified six distinct concern categories that any operationally meaningful governance instrument must address;⁴¹ Schuett, Reuel, and Carlier have specified the structural design of AI ethics boards required to translate principle into practice.⁴² The HGC³AE² Reference Architecture is one operational instantiation that responds to this body of work; its conformance criteria (§4) make the response auditable.

Three implications warrant explicit statement.

First, the architecture is decidable. An organization examining its own agentic operation can determine, by application of the cross-reference matrix and the conformance criteria, whether its deployment conforms or does not. The reference architecture is therefore an instrument for **structured self-assessment**, not only for greenfield deployment.

Second, the architecture is adaptable. The reference is not prescriptive about substrate choice (the federation pattern is one composition; others are admissible), about persona composition (the Non Sequitur reference uses a particular set of personas; others are

37 Wesley Gomes de Sousa et al., "How and Where Is Artificial Intelligence in the Public Sector Going? A Literature Review and Research Agenda," *Government Information Quarterly* 36, no. 4 (2019): 101392, <https://doi.org/10.1016/j.giq.2019.07.004>.

38 Gianluca Misuraca and Colin van Noordt, "Assessing the Public Policy-Cycle Framework in the Age of Artificial Intelligence," *Government Information Quarterly* 37, no. 4 (2020): 101509, <https://doi.org/10.1016/j.giq.2020.101509>.

39 U.S. Department of Defense, "DoD Adopts Ethical Principles for Artificial Intelligence," February 24, 2020, <https://www.defense.gov/News/Releases/Release/Article/2091996/dod-adopts-ethical-principles-for-artificial-intelligence/>.

40 Luciano Floridi, Josh Cowls, Monica Beltrametti, Raja Chatila, Patrice Chazerand, Virginia Dignum, Christoph Luetge, Robert Madelin, Ugo Pagallo, Francesca Rossi, Burkhard Schafer, Peggy Valcke, and Effy Vayena, "AI4People — An Ethical Framework for a Good AI Society: Opportunities, Risks, Principles, and Recommendations," *Minds and Machines* 28 (2018): 689–707, <https://doi.org/10.1007/s11023-018-9482-5>.

41 Brent Daniel Mittelstadt, Patrick Allo, Mariarosaria Taddeo, Sandra Wachter, and Luciano Floridi, "The Ethics of Algorithms: Mapping the Debate," *Big Data & Society* 3, no. 2 (2016), <https://doi.org/10.1177/2053951716679679>.

42 Jonas Schuett, Anka Reuel, and Alexis Carlier, "How to Design an AI Ethics Board," *AI and Ethics* (2024), <https://doi.org/10.1007/s43681-023-00409-y>.



admissible), or about operational workload mix (the federation pattern reflects mission-ready agentic operation as the primary workload; other workload mixes will shift the inflection points the architecture cites). What the reference *is* prescriptive about is the structural commitments at each layer — those are the conformance criteria.

Third, the architecture is generative. The framework's strength lies in the coupling of its parts, not in the parts themselves. Each part reinforces the others; each layer's outputs feed the layers it interfaces with; the loop closes. This is the property that the HGC³AE² framework's exponent captures and that the reference architecture as a whole exhibits.

The core claim of the Decalogy, restated at its close: **agentic systems fail not because they lack intelligence, but because they lack sufficient alignment architecture. The HGC³AE² Reference Architecture documented here is the architecture that, when applied, prevents that failure from being the default operating state.**

Returned to HGC³AE² letter by letter, integrating across all ten papers:

H — Human-governed. The Governance Layer (P1, P6) holds intent, constraints, and accountability. No version of the architecture is conformant without identifiable humans in this role.

G — Curated Context. The persistence plane (P7), the comprehension instrument's state-introspection function (P8), and the federation's authoritative persistence anchor (P9) jointly constitute the curated-context substrate.

C¹ — Cybersecurity. The architecture's security is governance-defined and substrate-spanning. The Managing Agentics Ops series develops the security and provenance surface in depth.⁴³

C² — Continuity. The cognitive node (P4), the persistence pillar (P4 Pillar 2), and the federation's resilience to substrate failure (P9) jointly preserve continuity across operational disruption.

C³ — Coherence. The unified governance envelope (P6), the comprehension instrument's drift detection (P8), and the federation's unified operating layer (P9) jointly enforce coherence across heterogeneity.

A — Agentically Engineered. The five-pillar node (P4), the Skipjack-conformant runtime (P5), the substrate planes (P7), and the comprehension instrument (P8) together constitute the agentically engineered execution layer.

E — Executed. Runtime execution is continuous (P4), enforced (P5), reported (P7 observability), self-comprehended (P8), and operationally validated (P9).

The exponent — the closure — is the architecture as a whole: operational telemetry from P9 feeds Governance Layer envelope updates per P6, which the federation executes against per P5, which is observed via P7's observability plane and self-comprehended via P8's instrument, completing the loop.

This paper closes the Mission-Ready AI Decalogy. The architecture's depth-of-operation will be developed in the Managing Agentics Ops Decalogy (the operational counterpart that walks each substrate, cost regime, and operational concern at length). The architecture's response to intent fidelity over operational time will be developed in the Intent Horizon Series,

⁴³ Forthcoming MAO-P9 *Security & Provenance* (v0.1-seed, in concurrent drafting; yks-build#102).



gate-blocked pending the closure of MR-Decalogy G3 and MAO-Decalogy G2.⁴⁴⁴⁵ The Edge Degraded Ops Undecalogy, parallel to MR-Decalogy, addresses the architecture's behavior under degraded conditions and provides the inheritance fixture for downstream coalition and accreditation work.⁴⁶

The reference architecture is complete. The implementation is up to the reader.

REFERENCES

[Bibliography compiled at draft v0.5. Cite-as-you-write population in progress; ≥12 peer-reviewed sources required at writing-step handoff per yks-canon/canon/bibliography.md §Citation discipline (locked 2026-05-11).]

— end v0.1-draft —

Drafting state: v0.1-rev1 capstone (chain v0.2 step 3, with PR-floor remediation rev1). MR-P1 shape applied: 6-section structure adapted for capstone (no Problem-Set section; replaced with components walk, cross-reference matrix, and conformance specification).

Citation count post-rev1: 12 peer-reviewed sources inline (Galster & Avgeriou QoSA-ISARCS 2011 · Angelov/Grefen/Greefhorst *Information and Software Technology* 2012 · Cloutier et al. *Systems Engineering* 2010 · Bass/Clements/Kazman *Software Architecture in Practice* · Kelly & Weaver DSN 2004 · Raji et al. FAT* 2020 · Floridi et al. *Minds and Machines* 2018 · Mittelstadt et al. *Big Data & Society* 2016 · Schuett/Reuel/Carlier *AI and Ethics* 2024 · de Sousa et al. *GIQ* 2019 · Misuraca & van Noordt *GIQ* 2020 · plus reference-architecture standards bundle TOGAF/SABSA/NIST-CSF counted under standards) — meets ≥12 PR floor independent of canon-inheritance “others” count. Canon refs P1-P9 retained with name+version per H-21.

44 Managing Agentics Ops Decalogy (in concurrent drafting; yks-build #91, #92, #93, #94, #100, #101, #102 + others).

45 Intent Horizon Series, gate-blocked behind MR G3 + MAO G2 (yks-build #111–#120).

46 Edge Degraded Ops Undecalogy promotion suite (yks-build #95, #96, #97, #103, #104, #105, #106, #107, #108, #109, #110).